

RESEARCH ARTICLE

Neural Estimation of Treatment Bridge Functions for Proximal Causal Inference

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ABSTRACT

The assumption of no unmeasured confounding is crucial for causal inference but is often unrealistic in observational studies. Recently proposed proximal causal inference offers a promising alternative to estimate the causal effect when informative proxy variables are available. An essential ingredient of proximal causal inference is solving integral equations based on bridge functions, such as outcome and treatment bridge functions. Distributional identification through the outcome bridge function often requires pointwise assumptions on the outcome distribution that may be difficult to verify or may fail in practice, prompting researchers to focus on the treatment bridge function, which imposes certain assumptions on the treatment. Various methods have been developed to estimate the treatment bridge function, yet data-adaptive and model-free approaches using deep learning are underexplored. In this work, we introduce DeepMMR, a novel method that leverages deep neural networks to estimate the treatment bridge function by minimizing a loss function derived from maximum moment restrictions. Then we exploit the estimated treatment bridge function to learn causal effects. Moreover, we provide theoretical consistency guarantees regarding our proposed framework. Through extensive simulations and real-world data analysis, we demonstrate that our method performs competitively in various settings.

1 | Introduction

Identifying causal effects is crucial for developing effective interventions and policies in the social, biological, and medical sciences. A commonly made assumption in causal effect estimation is that of no unmeasured confounding, which posits that a sufficiently rich set of covariates has been collected such that conditional exchangeability holds [1, 2]. However, this assumption may be unrealistic in observational studies, as it relies on researchers' ability to measure covariates relevant to all potential confounding mechanisms accurately. For example,

unmeasured confounders, such as parental smoking and phenotypes, could confound the intrauterine effect of maternal smoking on offspring outcomes [3–5]. Therefore, it is hoped that measured covariates can effectively act as proxies for the underlying confounding mechanisms, enabling us to recover true causal effects and draw valid conclusions.

Negative controls [6–9] utilize a specific set of covariates that share common causes with both the treatment and the outcome but exert no direct causal effect on either, aiming to detect confounding mechanisms and eliminate confounding bias. For a

recent review of negative control research and its applications, see [10, 11]. Although negative controls have long been utilized in laboratory experiments and epidemiological studies [12–15], the formal development of methods to debias causal effect estimates, known as proximal causal inference [16], has emerged more recently.

Proximal causal inference provides a powerful framework for estimating causal effects in the presence of unmeasured confounders, even though measured covariates may be imperfect proxies for the underlying confounding mechanisms. This framework requires researchers to classify a subset of measured covariates into three types: (i) variables that are common causes of both the treatment and the outcome; (ii) treatment-inducing confounding proxies; and (iii) outcome-inducing confounding proxies. Proxies of type (ii) are potential causes of the treatment, while proxies of type (iii) are potential causes of the outcome. Both types serve as proxies for the unmeasured common causes that simultaneously affect the treatment and the outcome. Generally speaking, covariates that are associated with unmeasured confounders but not causally related to either the treatment or the outcome can be classified as proxies of type (ii) or (iii). A key requirement for proximal causal inference is that both types of proxies (ii) and (iii) must be sufficiently informative about the unobserved confounders. This condition ensures the existence of bridge functions, which are essential ingredients of the framework and facilitate causal effect identification without directly capturing unmeasured confounders or estimating their distribution.

The concept of identifying causal effects through an outcome bridge function was introduced by [8], which is characterized by a Fredholm integral equation. To estimate the outcome bridge function, Tchetgen Tchetgen et al. [16] introduced proximal g-computation, a generalization of Robins' g-computation, and a two-stage least squares estimator that relies on a linear bridge function and completeness assumptions. Mastouri et al. [17] extended the work and developed a two-stage kernel ridge regression method, which remains applicable when the full sample distribution is unavailable. Further advancing this work, Xu et al. [18] incorporated neural networks into the two-stage regression framework, relying on completeness conditions and enabling deep adaptive feature learning.

Beyond regression-based approaches, another line of research formulates estimation as an optimization problem under moment restrictions. Muandet et al. [19] introduced the Maximum Moment Restriction (MMR) framework, which transforms conditional moment restrictions into unconditional ones and constrains the function space within the unit ball of a reproducing kernel Hilbert space (RKHS) [20, 21]. Building upon this idea, Mastouri et al. [17] proposed the Proxy Maximum Moment Restriction (PMMR) method for estimating bridge functions, utilizing U-statistics and V-statistics [22] to formulate an empirical risk minimization objective. Kompa et al. [23] introduced Neural Maximum Moment Restriction (NMMR), which incorporates neural networks into this framework, thereby enhancing its adaptability to complex data distributions.

However, the above identification strategies for the causal estimates, which rely on the outcome distribution, often necessitate

a pointwise assumption on the outcome bridge function that may be difficult to verify or may fail in practice. Consequently, researchers propose a different identification strategy through the so-called treatment bridge function. Cui et al. [24] proposed evaluating causal effects through the treatment bridge function, which is obtained by solving another Fredholm integral equation. Various methods have been developed to estimate the treatment bridge function. Cui et al. [24] proposed a semiparametric approach that leverages the Generalized Method of Moments (GMM) to estimate parameters, with the treatment bridge function modeled in a logit form. Kallus et al. [25] introduced a minimax learning approach that avoids the completeness condition and relies solely on the existence of bridge functions. This method accommodates various function classes, including RKHSs and neural networks. Similarly, Ghassami et al. [26] also developed a minimax framework that employs doubly robust moment functions to construct estimation equations applicable to a broader range of function spaces.

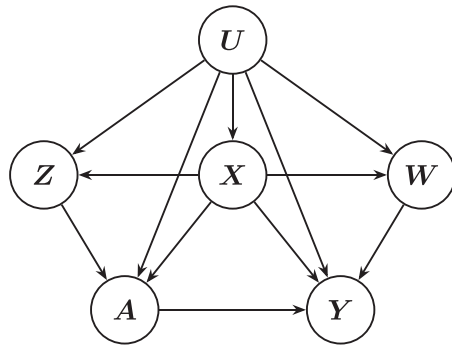
Limited research has considered data-adaptive and model-free approaches for estimating the treatment bridge function, leaving the potential of deep learning underexplored. In this work, we propose DeepMMR, a novel method that utilizes deep neural networks to estimate the treatment bridge function by minimizing a loss function based on maximum moment restrictions. Once the estimated treatment bridge function is obtained, the average treatment effect can be calculated. Our approach operates under several sufficient assumptions, and the consistency of our estimator is ensured. Our method achieves competitive performance in simulations and is applicable to real-world data analysis.

The remainder of this paper is structured as follows: Section 2 introduces the framework of proximal causal inference and formulates the problem of interest. In Section 3, we propose a neural network-based estimator for the treatment bridge function and the causal effect. Consistency regarding the estimator is established in Section 4. Section 5 evaluates the performance of our method compared to other approaches through extensive simulations. We demonstrate the practical applicability of our method using the right heart catheterization dataset in Section 6. Finally, Section 7 concludes the work with a discussion of our findings and potential directions for future research. Proofs and additional results are provided in the Appendix. Before proceeding, we summarize the key notations used throughout the paper for clarity.

Notation. The uppercase letters (e.g., A), calligraphic letters (e.g., $\mathcal{A}$), and lowercase letters (e.g., a) represent random variables, measurable spaces, and realizations of random variables, respectively. The symbols $\mathbb{P}$ and $\mathbb{E}$ denote the probability distribution and the expectation of a random variable. The indicator function is denoted by $\mathbb{I}$ and $X \perp Y$ denotes independence between random variables X and Y . Hats (e.g., $\hat{\phi}$) indicate estimators. Let $\mathcal{D} = \{d_i\}_{i=1}^n$ denote a dataset consisting of n independent and identically distributed samples.

2 | Problem Formulation

Causal inference aims to estimate the effect of a treatment A on an outcome Y , using either observational data or randomized experiment results. In this work, we focus on the



Variable:

- U : Unmeasured confounders
- Z : Treatment-inducing confounding proxies
- X : Observed confounders
- W : Outcome-inducing confounding proxies
- A : Treatment
- Y : Outcome

FIGURE 1 | A causal directed acyclic graph of proximal causal inference.

binary treatment setting, where $A \in \{0, 1\}$. A central estimand of interest is the average treatment effect (ATE), defined as $\varphi = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$, which quantifies the expected difference in the outcome Y caused by changing the treatment A from 0 to 1 in the population. In the absence of confounding, the ATE is equal to the difference in conditional means $\mathbb{E}[Y|A = 1] - \mathbb{E}[Y|A = 0]$. However, it generally does not hold in observational studies due to confounding.

To control for confounding, investigators typically measure a set of covariates L and incorporate them into the analysis as adjustment variables. When these covariates are sufficiently informative, the assumption of exchangeability holds, that is, $Y(a) \perp A | L$ for $a = 0, 1$. This assumption relies on the investigators' ability to accurately observe all variables that account for potential confounding. However, in practice, it is often infeasible to collect a sufficient set of confounders. Instead, observed covariates typically act as proxies for the underlying unmeasured confounders. Proximal causal inference leverages these covariates as proxies for confounding mechanisms, enabling identification of the causal effect in the presence of unmeasured confounding.

2.1 | Proximal Causal Inference

As previously noted, unobserved confounders U present a significant challenge to the estimation of true causal effects. Proximal causal inference provides an alternative approach by utilizing observed variables that serve as proxies for latent confounders. In this framework, covariates are classified into three types: treatment-inducing confounding proxies Z , outcome-inducing confounding proxies W , and observed confounders X that influence both the treatment A and the outcome Y . Specifically, the treatment-inducing confounding proxies are potential causes of the treatment, and the outcome-inducing confounding proxies are potential causes of the outcome. Figure 1 illustrates a representative directed acyclic graph (DAG) that satisfies the identification conditions of proximal causal inference.

It is noted that the observed common causes X help reduce residual confounding between A and Y , while the confounding proxies serve to link the unobserved confounder U to the treatment A and the outcome Y , that is, Z and W act as bridges that mediate the influence of U without requiring direct observation of U . The observed conditional distributions involving these proxies contain sufficient information to recover the impact of unmeasured

confounding. Under appropriate assumptions, proximal causal inference leverages these conditional distributions—by modeling the confounding mechanism through Fredholm integral equations of the first kind—to identify the causal effect of A on Y without observing U . To ground this framework in practice, we next present a real-world example that involves proxies and illustrates how the proximal causal inference framework can be applied, followed by a discussion of the key assumptions required for identification.

An illustrative application of proximal causal inference arises in evaluating the causal effect of flu shots A on influenza hospitalization Y among the elderly [6, 10]. In this context, unmeasured health-seeking behavior, often difficult to capture, confounds the causal effect. Nevertheless, there are valid proxies for the unmeasured confounders U . For instance, the history of annual wellness visits, which reflects an individual's willingness to engage in healthy behaviors but is not associated with influenza hospitalization, can serve as a treatment-inducing confounding proxy Z . Similarly, injury/trauma hospitalization, which is associated with health-seeking behavior but not influenced by flu shots, can serve as an outcome-inducing confounding proxy W . Lifestyle choices, such as exercising and nutritional habits, can be considered as observed confounders X . With these proxies, we could recover the impact of the unmeasured confounding and identify the true causal effect. Figure 2 illustrates the above discussion and serves as the real-world counterpart to the general DAG shown in Figure 1. Additional real-world applications can be found in [9, 27–29].

To enable valid causal inference in real-world applications like the one described above, the proximal inference framework requires certain key assumptions to be satisfied.

Assumption 1. (Structural). 1. Consistency: $Y(a) = Y$ almost surely when $A = a$.

2. Positivity: $0 < \Pr(A = a | U, X) < 1$ almost surely, $a = 0, 1$.

3. Conditional independence: $Y \perp Z | A, U, X$ and $W \perp (A, Z) | U, X$.

4. Conditional randomization: $Y(a) \perp A | U, X$ for $a = 0, 1$.

Condition (1) guarantees that for individuals receiving treatment $A = a$, the observed outcome Y is the realization of the potential

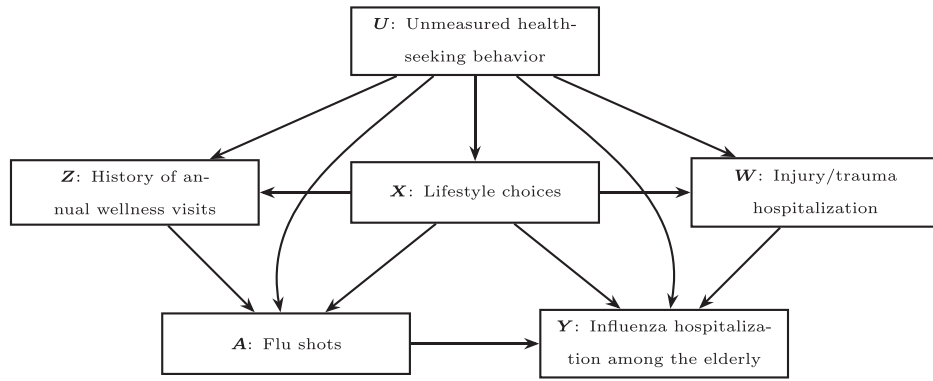


FIGURE 2 | Causal diagram of the effect of flu shots on influenza hospitalization among the elderly under the proximal causal inference framework.

outcome $Y(a)$. Condition (2) ensures that, within every stratum defined by (U, X) , there is a positive probability of receiving any treatment level. Condition (3) clarifies the requirements of proxies: conditional on (A, U, X) , there is no causal effect of Z on Y , and conditional on (U, X) , there is no causal effect of (A, Z) on W . Condition (4) requires latent exchangeability. Figure 1 provides a graphical representation of Conditions (3) and (4). In addition to these structural assumptions, a completeness assumption is essential for the identification of causal effects.

Assumption 2. (Completeness). For any square integrable function g and for any $a \in \mathcal{A}$, $x \in \mathcal{X}$, $\mathbb{E}[g(U) | A = a, W, X = x] = 0$ almost surely if and only if $g(U) = 0$ almost surely.

The completeness assumption rules out conditional independence of U and W given A and X . It means that proxies W have sufficient variability relative to U . In the case of categorical U and W with a number of categories d_u and d_w respectively, Assumption 2 implies $d_w \geq d_u$, which means the proxies W must have at least as many categories as U [16, 24]. We further assume the following existence of a treatment bridge function [24].

Assumption 3. (Treatment bridge function). Suppose there exists a treatment bridge function $q(a, x, z)$ that solves the following integral equation:

$$\mathbb{E}[q(A, X, Z) | A = a, W, X] = \frac{1}{f(A = a | W, X)}, \quad a = 0, 1 \quad (1)$$

where f denotes the conditional density.

The treatment bridge function q defined in Equation (1) is particularly motivated by its role in identifying causal effects. In scenarios without unmeasured confounding, causal effects are identified through inverse weighting by the propensity score $f(A|X)$. However, when unmeasured confounders are present, the conditional density $f(A|U, X)$ becomes non-identifiable directly from data. Following [24] with Assumptions 1–3, the identity $\mathbb{E}[q(A, X, Z) | A = a, U, X] = 1/f(A = a | U, X)$ holds, which makes the function q a valid tool for identifying causal quantities even in the presence of unmeasured confounding. Once the treatment bridge function q is obtained, the ATE can be identified via the following equation:

$$\varphi = \mathbb{E}[(-1)^{1-A} q(A, X, Z) Y]$$

In addition, the treatment bridge function q enables the identification of quantile treatment effects (QTE). For $\tau \in (0, 1)$, the τ -th quantile treatment effect [30, 31] is given by

$$\psi(\tau) = Q_{1,\tau} - Q_{0,\tau}$$

where $Q_{a,\tau} = \inf\{y : \Pr[Y(a) \leq y] \geq \tau\}$, $a = 0, 1$. By a simple calculation, the τ -th quantile of the potential outcome $Q_{a,\tau}$ can be identified via the following equation

$$\mathbb{E}[\mathbb{I}(Y(a) \leq Q_{a,\tau})] = \mathbb{E}[\mathbb{I}(A = a) q(A, X, Z) \mathbb{I}(Y \leq Q_{a,\tau})] = \tau \quad (2)$$

Another important bridge function used in the literature for causal identification is the outcome bridge function h , as introduced by [8] and discussed in Section 1. However, the identification of quantile treatment effects requires a pointwise assumption on the outcome bridge function, that is, h is a function of y . Therefore, we focus on the treatment bridge function q instead. To streamline our presentation, we will focus on φ , and the proposed method can be naturally extended to $\psi(\tau)$.

Accurate estimation of causal effects hinges on first estimating the bridge function q , which satisfies Equation (1), a Fredholm integral equation of the first kind [32] known to be an ill-posed inverse problem. The closed-form solution of the equation is not always available, and solving it can be computationally intensive and numerically unstable. Even a small amount of uncertainty in the input may lead to excessive variability in the solution. Taking these into consideration, we introduce an estimation framework for the bridge function that ensures stable and reliable results. In Section 2.2, we introduce the target risk function that underpins our approach. In Section 3, we develop a neural network-based method that leverages implicit regularization. The resulting bridge function estimates are then used to approximate the causal effect.

2.2 | Maximum Moment Restriction

Equation (1) can be rewritten as

$$\mathbb{E}[\mathbb{I}(A = a) q(A, X, Z) - 1 | W, X] = 0, \quad a = 0, 1 \quad (3)$$

which serves as a conditional moment restriction (CMR). Then we transform the above CMR into a continuum of unconditional

moment restrictions (UMRs) [19], that is,

$$\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)p(W, X)] = 0, \quad \forall p \in \mathcal{P} \quad a = 0, 1 \quad (4)$$

where $\mathcal{P}$ denotes the set of all measurable functions from $\mathcal{W} \times \mathcal{X}$ to $\mathbb{R}$. Although this transformation captures all information about the original CMR and can somewhat simplify computations, it also introduces an infinite number of UMRs corresponding to the infinite number of measurable functions in $\mathcal{P}$. Since Equation (4) must hold for all $p \in \mathcal{P}$, it can be reformulated using the supremum operator as follows:

$$\sup_{p \in \mathcal{P}} (\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)p(W, X)])^2 = 0, \quad a = 0, 1 \quad (5)$$

This is the previously introduced Maximum Moment Restriction (MMR) method. Although this procedure reduces the dimensionality of UMRs, the sup operator complicates the optimization of Equation (5). To further address this challenge, we next consider function spaces with specific structures, focusing on the reproducing kernel Hilbert space defined in Assumption 4.

Assumption 4. Let $\mathcal{P}_k$ be a Hilbert space defined on $(\mathcal{W}, \mathcal{X})$ with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. The space $\mathcal{P}_k$ is a reproducing kernel Hilbert space (RKHS) equipped with a reproducing kernel $k : (\mathcal{W}, \mathcal{X}) \times (\mathcal{W}, \mathcal{X}) \rightarrow \mathbb{R}$, which satisfies the following properties: (i) $k((w, x), (\cdot, \cdot)) \in \mathcal{P}_k$, for all $(w, x) \in (\mathcal{W}, \mathcal{X})$, and (ii) (reproducing property) $p(w, x) = \langle p, k((w, x), (\cdot, \cdot)) \rangle$ for all $p \in \mathcal{P}_k$ and $(w, x) \in (\mathcal{W}, \mathcal{X})$.

Let p denote a function in an RKHS $\mathcal{P}_k$ equipped with a reproducing kernel k . Following [19], when the functions are restricted to a unit ball in $\mathcal{P}_k$, the treatment bridge function q can be estimated by minimizing the risk function below,

$$R_{\mathcal{P}_k}^a(q) = \sup_{p \in \mathcal{P}_k, \|p\| \leq 1} (\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)p(W, X)])^2, \quad a = 0, 1 \quad (6)$$

And we assume that the minimizer of $R_{\mathcal{P}_k}^a(q)$ satisfies the following assumption.

Assumption 5. Consider the function space $\mathcal{Q}$ and $q^* \in \arg \min_{q \in \mathcal{Q}} R_{\mathcal{P}_k}^a(q)$. Then for any $q' \in \mathcal{Q}$ with $\mathbb{E}[|q'(A, X, Z)|] < \infty$, $\mathbb{E}[(\mathbb{I}(A = a)(q'(A, X, Z) - q^*(A, X, Z)) | W, X)] = 0$ implies $q' = q^*$.

Assumption 5 ensures the uniqueness of the optimal function q^* and provides a sufficient condition for consistency. It is evident that restricting the function class to a unit ball in the RKHS, instead of considering all measurable functions, significantly simplifies the computation. Furthermore, we can leverage the reproducing property of the RKHS to reformulate the equation using the kernel function k , as demonstrated in Lemma 1.

Lemma 1. Assume that $\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)^2 k((W, X), (W, X))] < \infty$, and denote by (A', W', X', Y', Z') an independent copy of the random variables (A, W, X, Y, Z) . Then, we have $R_{\mathcal{P}_k}^a(q) = R_k^a(q)$ for $a = 0, 1$, where $R_k^a(q)$ is defined as:

$$R_k^a(q) = \mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1) \times (\mathbb{I}(A' = a)q(A', X', Z') - 1)k((W, X), (W', X'))]$$

Lemma 1 builds upon the insights from Lemma 1 in [33] and Lemma 2 in [17]. Specifically, it drops the sup operator in Equation (6) and makes the risk $R_k^a(q)$ depend on the choice of the kernel k . The preceding discussion draws a rough link to connect Equation (3) with $R_k^a(q)$. Hereafter, we present a formal theorem for linkage, assuming that the reproducing kernel k is integrally strictly positive definite, as specified in Assumption 6.

Assumption 6. The kernel k is continuous, bounded, and integrally strictly positive definite (ISPD). That is, for any square integrable function g with $0 < \|g\|_2^2 < \infty$, the following holds: $\iint g(\xi)k(\xi, \xi')g(\xi')d\xi d\xi' > 0$, where $\xi, \xi' \in (\mathcal{W}, \mathcal{X})$.

Theorem 1. Under Assumptions 4 and 6, for any real-valued measurable function q , $R_k^a(q) = 0$, if and only if $\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1) | W, X] = 0$ holds $\mathbb{P}_{W, X}$ -almost surely.

According to Theorem 1, if we find a treatment bridge function q such that $R_k^a(q) = 0$, we could obtain a solution that $\mathbb{P}_{W, X}$ -almost surely solves Equation (1). At the same time, Theorem 1 shows that although different ISPD kernels may yield distinct representations and numerical approximations in practice, the true underlying bridge function remains unchanged. Hence, any ISPD kernel can be used without affecting the solution, providing greater flexibility in kernel selection.

To end this part, we now state the main problem, that is,

$$\min_q R_k^a(q)$$

with a properly selected k for $a = 0, 1$. And we will discuss how to use dataset $\mathcal{D} = \{d_i\}_{i=1}^n = \{(a_i, w_i, x_i, y_i, z_i)\}_{i=1}^n$ to solve this in the next section.

3 | Risk Minimization via Neural Networks

In this section, we start by approximating the risk function via the empirical risk. We then employ a deep neural network to estimate q by minimizing the risk function. Then, the ATE is estimated with the assistance of $\hat{q}$. Additionally, we provide a brief complexity analysis of the proposed method.

3.1 | Empirical Risk Functions

Given the data $\mathcal{D}$, to approximate $R_k^a(q)$, we use empirical risks in the form of U-statistics and V-statistics [22] as shown below,

$$\hat{R}_{k,U}^a(q) = \frac{1}{n(n-1)} \sum_{i,j=1, i \neq j}^n (\mathbb{I}_i q_i - 1)(\mathbb{I}_j q_j - 1)k_{ij} \quad (7)$$

$$\hat{R}_{k,V}^a(q) = \frac{1}{n^2} \sum_{i,j=1}^n (\mathbb{I}_i q_i - 1)(\mathbb{I}_j q_j - 1)k_{ij} \quad (8)$$

where $\mathbb{I}_i := \mathbb{I}(a_i = a)$, $q_i := q(a_i, x_i, z_i)$, and $k_{ij} := k((w_i, x_i), (w_j, x_j))$. Both forms, namely $\hat{R}_{k,U}^a(q)$ and $\hat{R}_{k,V}^a(q)$, can be used for a consistent estimation of q . To tell the difference, a U-statistic considers tuples of independent samples, that is, pairs $(i, j) \in [n] \times [n]$ with $i \neq j$. In contrast, a V-statistic includes

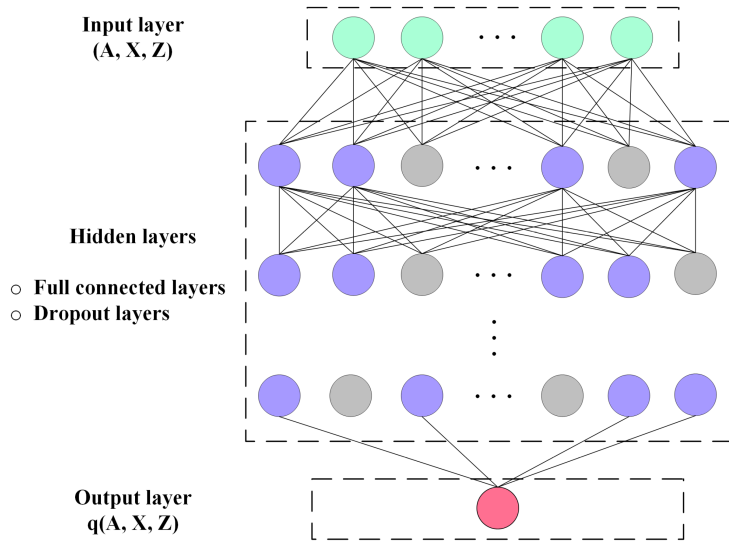


FIGURE 3 | A diagram of the neural network architecture.

all tuples (i, j) , and for pairs where $i = j$, d_i and d_j are dependent. Thus, $\hat{R}_{k,U}^a(q)$ is a minimum-variance unbiased estimator of $R_k^a(q)$, whereas $\hat{R}_{k,V}^a(q)$ is a biased estimator [17, 22]. However, in practice, the V-statistic utilizes full samples and may sometimes yield a better estimator than that of the U-statistic.

As we proceed to construct the neural architecture, we also provide a representation in vector and matrix forms. Define the column vectors $\mathbf{I} := (\mathbb{I}_1, \mathbb{I}_2, \dots, \mathbb{I}_n)^T$, $\mathbf{q} := (q_1, q_2, \dots, q_n)^T$, and $\mathbf{1} := (1, 1, \dots, 1)^T$. Let $K = [k_{ij}]_{i,j=1}^n \in \mathbb{R}^{n \times n}$ be a symmetric kernel matrix, where $k_{ij} = k((w_i, x_i), (w_j, x_j))$ is defined by the reproducing kernel k . With these notations, Equations (7) and (8) can be rewritten as:

$$\hat{R}_{k,U(V)}^a(q) = (\mathbf{I}\mathbf{q} - \mathbf{1})^T K_{U(V)} (\mathbf{I}\mathbf{q} - \mathbf{1}) \quad (9)$$

where $K_{U(V)}$ is determined by the kernel matrix K . Specifically, K_U is derived by removing the diagonal elements of K and normalizing it with $n(n-1)$, and $K_V = K/n^2$. Since the choice of kernel does not affect the solution, we adopt the RBF kernel throughout the remainder of the paper.

3.2 | Neural Architecture and Approximation

To minimize the empirical risk and estimate the treatment bridge function $q(A, X, Z)$, we employ a fully connected feedforward neural network. The architecture maps the input features (A, X, Z) directly to a continuous estimate of q through multiple hidden layers with ReLU activation functions, allowing the model to capture complex nonlinear relationships in the data. To mitigate overfitting, dropout layers are strategically included, and the final estimate of q is generated by a linear output layer, producing continuous-valued predictions. Figure 3 presents a schematic diagram illustrating our neural network design.

As previously discussed, the ill-posed nature of Fredholm integral equations presents significant challenges in stably and reliably estimating the treatment bridge function q . To overcome these,

despite the empirical risk functions, we also integrate a L_2 regularization term into the objective function, penalizing excessively complex solutions and thereby enhancing numerical stability. Following [23], the resulting optimization problem is defined as:

$$\min_q (\mathbf{I}\mathbf{q} - \mathbf{1})^T K_{U(V)} (\mathbf{I}\mathbf{q} - \mathbf{1}) + \lambda \|\mathbf{q}\|_2^2 \quad (10)$$

where λ balances goodness of fit and model complexity. Furthermore, as illustrated in the architectural design, stochastic optimization and dropout layers are incorporated as additional regularization techniques, enhancing robustness to noise and improving the stability of the estimator.

To effectively train the neural network and facilitate efficient optimization, we follow deep learning practices [34–36] when configuring key training parameters such as batch size and optimizer type. Building upon the network architecture described above, we conduct hyperparameter tuning by adjusting the number of hidden layers, the number of neurons per layer, the learning rate, the dropout rate, and the L_2 weight decay. This procedure is implemented using the `gp_minimize` function from the `scikit-optimize` Python library.

Once the estimator $\hat{q}_{U(V)}$ is obtained by solving (10), the ATE is finally estimated via

$$\hat{\varphi}_{U(V)} = \frac{1}{n} \sum_{i=1}^n (-1)^{1-a_i} \hat{q}_{U(V)}(a_i, x_i, z_i) y_i$$

While the kernel matrix is central to our estimation framework, its computations constitute the primary bottleneck in terms of both time and memory. Specifically, constructing the kernel matrix to capture pairwise relationships incurs a time and space complexity of $\mathcal{O}(n^2)$, where n denotes the number of samples. As n increases, both the computational burden and memory usage grow quadratically, making the kernel matrix a key obstacle to scalability.

To mitigate this limitation, several optimization strategies can be considered. These include low-rank kernel approximations

TABLE 1 | Parameter settings for different scenarios.

Scenario	$b_1(X)$	$b_2(X)$	$b_3(X)$	b_a	b_w
S1	$4 + X_0 - 1 - X_1 + 1 $	X^T	$\sin(X_0) - \cos(X_1)$	-1.5	2
S2	$4 - 4X_0X_1$	X^T	X_0	-3	2
S3	$4 - 2X_0 + X_1$	$(0.25, 0.25)^T$	X_1	-3	2
S4	$2 + \cos(X_0) - \sin(X_1)$	$(0.25, 0.25)^T$	$3 \exp(X_1) - \exp(X_0)$	-3	1
S5	$4 + \exp(X_0) - \exp(2X_1)$	$(0.25, 0.25)^T$	$2 X_1 + 1 - 3 X_0 - 1 $	-2.5	1
S6	$4 + \sqrt[3]{X_0} - 3\sqrt[3]{X_1}$	X^T	X_1^2	-1.5	1

and block-wise matrix computations, which aim to reduce the computational burden while preserving the expressive power of kernel-based methods. In our implementation, we adopt a mini-batch training scheme to reduce computational load and improve efficiency, with further details provided in Appendix E.

4 | Theoretical Guarantees

In this section, we provide theoretical guarantees for our method. We first establish the consistency of the estimated treatment bridge functions, followed by the consistency of the estimated ATE. The proofs are primarily based on [33], with full details and additional results provided in Appendix A.

Theorem 2. (Consistency of $\hat{q}_{U(V)}$). Let $q^* : \mathcal{A} \times \mathcal{X} \times \mathcal{Z} \rightarrow \mathbb{R}$ satisfy $\mathbb{E}[\mathbb{I}(A = a)q^*(A, X, Z) - 1|W, X] = 0$ $\mathbb{P}_{W, X}$ -almost surely. Assume that $\mathbb{E}[|q(A, X, Z)|] < \infty$ ($\mathbb{E}[\sup_{q \in \mathcal{Q}} |q(A, X, Z)|^2] < \infty$), $\mathcal{Q}$ is compact and convex, Assumptions 5 and 6 hold, $\Lambda(q)$ is a bounded regularizer and $\lambda \xrightarrow{P} 0$. Then $\hat{q}_{U(V)} \xrightarrow{P} q^*$.

Theorem 2 demonstrates that under several conditions, the estimated treatment bridge functions $\hat{q}_U$ and $\hat{q}_V$ consistently converge to the true treatment bridge function q^* as the sample size increases. Building upon this, we now establish the consistency of the ATE estimators $\hat{\varphi}_U$ and $\hat{\varphi}_V$ in Theorem 3.

Theorem 3. (Consistency of $\hat{\varphi}_{U(V)}$). Assume that conditions of Theorem 2 hold and the outcome Y is bounded, $\hat{\varphi}_{U(V)} \xrightarrow{P} \varphi$.

Remark. It should be noted that although our theoretical results establish consistency of the estimated treatment bridge function $\hat{q}$ and the ATE estimator $\hat{\varphi}$ in large samples, practical challenges such as estimator instability persist in finite samples. Specifically, regions where the conditional density $f(A = a|W, X)$ is close to zero lead to large inverse probability weights and consequently high estimator variance. In practice, our framework addresses this issue by incorporating an explicit regularization term that penalizes excessively large solutions, along with utilizing the neural network's global approximation capability to achieve smoother estimates.

5 | Simulation

5.1 | Data Generation

To assess the performance of different methods for estimating causal effects, we utilize a data generation mechanism based

TABLE 2 | A list of proximal causal inference methods.

Method	Description
PMMR [17]	Minimization optimization with kernel decomposition
GMM [24]	Semiparametric model regression in a logit form
MinMax [26]	Minimax optimization with a closed-form solution
DeepMMR-U (ours)	Neural network with a regularized U-Statistic risk
DeepMMR-V (ours)	Neural network with a regularized V-Statistic risk

on the one outlined in [24, 37]. More details can be found in Appendix B.

We design six different scenarios, varying $b_1(X)$, $b_2(X)$, $b_3(X)$, b_a , and b_w , with the parameter settings summarized in Table 1. For each scenario, we generate training datasets $\{(a_i, w_i, x_i, y_i, z_i)\}_{i=1}^n$ with sample sizes of $n = 2000$ and $n = 4000$. Additionally, we evaluate the ground truth ATE φ using datasets generated from the same mechanism, each with a sample size of 10^6 .

5.2 | Evaluation

To evaluate the performance of the models, we compute the ratio of the estimated causal effect $\hat{\varphi}$ to the true causal effect φ . This metric directly measures how well the estimated effects align with the true causal effects, offering a clear evaluation of the accuracy of estimation.

We compare the performance of our proposed methods (DeepMMR-U and DeepMMR-V) with other approaches that estimate causal effects using bridge functions. The methods are presented in Table 2, and the code can be found at <https://github.com/Bingxi-Zhang/DeepMMR>.

5.3 | Results

All experiments were conducted using Python 3.12.5 and PyTorch 2.4.1+cpu. Hyperparameter tuning details are provided in Appendix C. Each simulation was replicated 100 times. Figures 4 and 5 illustrate the experimental results of

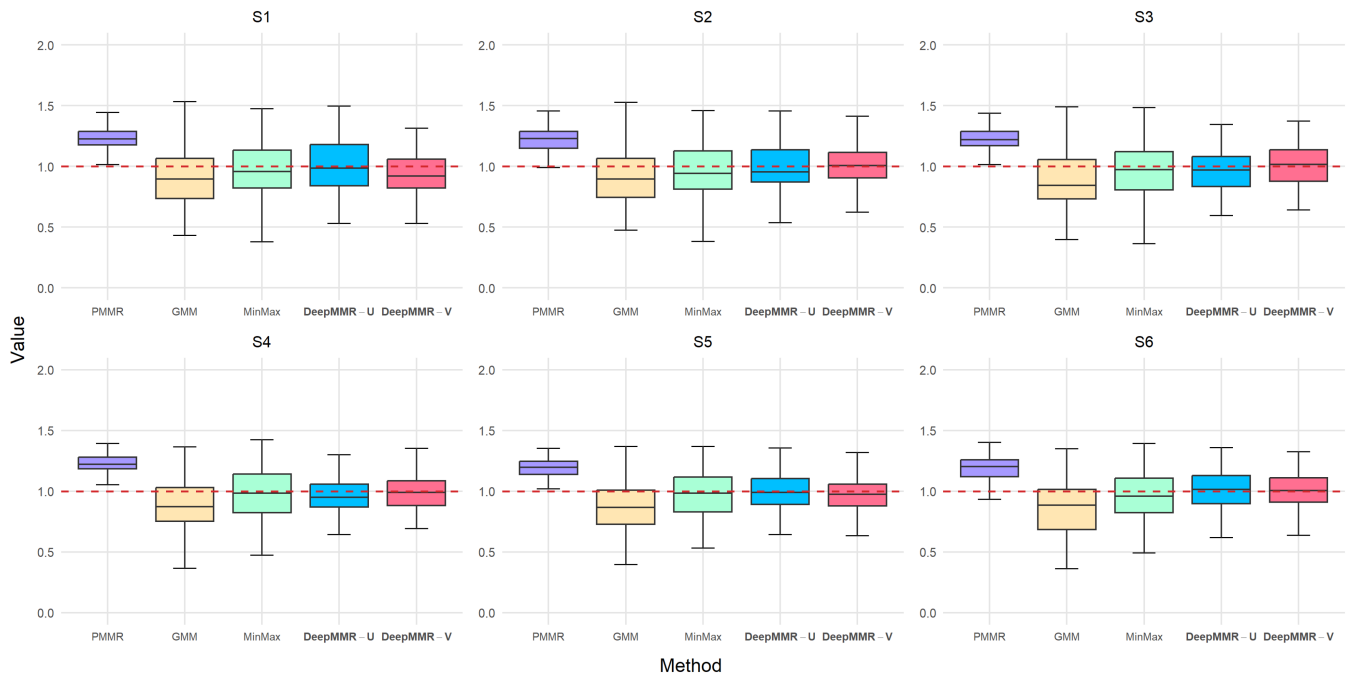


FIGURE 4 | Experimental results ($n = 2000$): The performance of different methods (PMMR, GMM, MinMax, DeepMMR-U, DeepMMR-V) across the various data generation scenarios (S1–S6).

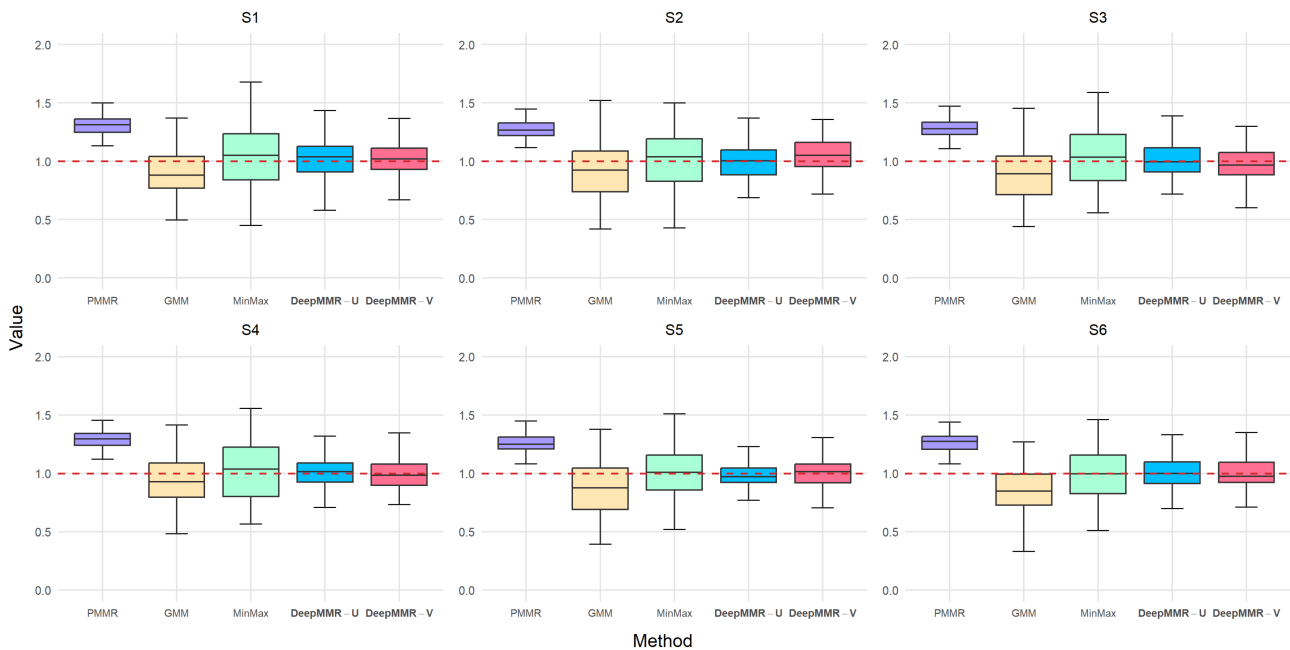


FIGURE 5 | Experimental results ($n = 4000$): The performance of different methods (PMMR, GMM, MinMax, DeepMMR-U, DeepMMR-V) across the various data generation scenarios (S1–S6).

various methods across different scenarios. For detailed results, including means and standard deviations, see Tables D1 and D2 in Appendix D.

From these graphs, it is evident that MinMax, DeepMMR-U, and DeepMMR-V consistently outperform other methods across all scenarios (S1–S6), demonstrating superior adaptability in handling complex data structures. Among these, DeepMMR-U and DeepMMR-V achieve superior performance compared to

MinMax, which serves as a reliable estimator. Between our proposed methods, DeepMMR-U occasionally performs better than DeepMMR-V, benefiting from its use of independent sample pairs. Although DeepMMR-V introduces bias, it demonstrates stable performance in certain settings, as seen in scenario S6.

When comparing results across datasets of different sizes, we observe that increasing the sample size from 2000 to 4000 generally enhances the performance of DeepMMR-U and

TABLE 3 | Average treatment effect estimates (standard errors) and 95% confidence intervals.

	DeepMMR-U	DeepMMR-V
ATE (SEs)	−2.606 (0.292)	−2.171 (0.112)
95% CIs	(−3.177, −2.034)	(−2.391, −1.950)

DeepMMR-V. This improvement indicates their ability to efficiently process larger datasets and demonstrates their suitability for complex, data-intensive tasks. See the performance of DeepMMR-U and DeepMMR-V in S4 and S6 for illustration.

6 | Real Data Analysis

In this section, we illustrate the proposed estimators of the ATE using a data application considered by [16, 24, 37, 38]. Specifically, we utilize the right heart catheterization (RHC) dataset from the Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments (SUPPORT) to estimate the causal effect of RHC during the first 24 h of care in the intensive care unit (ICU) on survival time up to 30 days [39].

The dataset consists of 5735 individuals, among whom 2184 received RHC treatment and 3551 were under control. Among these patients, 3817 survived, whereas 1918 died within 30 days. The study collected extensive patient information, encoded in various covariates. For more details on the covariates, see [40]. In our study, the treatment A indicates whether a patient received RHC during the first 24 h of care in the ICU, and the outcome Y is the number of days between admission and either death or censoring at 30 days. Meanwhile, 10 variables assessing the patients' overall physiological status were measured based on blood tests conducted within the first 24 h in the ICU. These variables, prone to substantial measurement error, offer only a single snapshot of the patient's underlying physiological state, and thus may serve as potential confounding proxies. Following the settings in [16, 24, 37], we choose $Z = (\text{paf1}, \text{paco21})$ and $W = (\text{ph1}, \text{hema1})$. And we assign 20 variables to X , all of which show significant associations with both the treatment A and the outcome Y . Detailed information is provided in Appendix F.

Table 3 presents the point estimates and 95% confidence intervals for the ATE. Our methods yield causal effect estimates that are consistent with previous studies employing the bridge functions h or q [24, 27]. This indicates that the RHC treatment may have a negative impact on the 30-day survival rates of critically ill patients admitted to the ICU.

7 | Discussion

In this paper, we propose a novel deep learning method for estimating treatment bridge functions and evaluating causal effects in the presence of unmeasured confounding. Building on the framework of proximal causal inference, our approach leverages deep neural networks to solve optimization problems derived from moment restrictions, offering a data-adaptive and model-free estimation strategy. By incorporating effective

regularization techniques, the method mitigates the ill-posedness inherent in Fredholm integral equations of the first kind and enhances the stability of estimation in finite samples.

Despite its practical effectiveness, the proposed method encounters notable computational challenges due to the explicit use of kernel matrices. Specifically, the computation of the kernel matrix incurs a time and space complexity of $\mathcal{O}(n^2)$, where n denotes the number of samples. To alleviate this computational burden, we adopt a mini-batch strategy during training, which significantly reduces memory usage and improves runtime efficiency. As demonstrated in Table E1 in Appendix E, this approach yields substantial gains in both computational speed and resource utilization. To further address this limitation, future work may explore scalable kernel approximation techniques—such as random Fourier features [41] or Nyström methods [42, 43]—to approximate the kernel matrix and reduce complexity without compromising estimation accuracy.

Our method employs RKHS as the function class for p , which facilitates both theoretical analysis and practical implementation. Nevertheless, alternative function classes such as neural networks offer greater flexibility and representational power. However, adopting such alternatives introduces new challenges, as the resulting minimax formulation becomes non-convex, which may require more advanced optimization strategies (e.g., simultaneous Adam [44] or optimistic Adam [45]). Addressing these challenges presents an important avenue for future research.

Another promising extension lies in the development of doubly/multiply robust deep learning methods for proximal causal inference. By incorporating both the treatment and outcome bridge functions, as demonstrated in [24, 25, 46, 47], one may enhance the robustness and efficiency of causal effect estimation under unmeasured confounding.

Author Contributions

Bingxi Zhang: methodology, implementation, visualization, writing. **Tao Shen:** methodology, validation, writing. **Yifan Cui:** methodology, writing, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in GitHub at <https://github.com/Bingxi-Zhang/DeepMMR>.

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Appendix A

Proofs

Proof of Equation (2)

Proof. First, we prove the following equation:

$$\begin{aligned} \mathbb{E}[\mathbb{I}(Y \leq y) | A = a, U, X = x] \\ = \mathbb{E}[\mathbb{I}(A = a)q(A, X, Z)\mathbb{I}(Y \leq y) | U, X = x] \end{aligned} \quad (\text{A1})$$

We have

$$\begin{aligned} \mathbb{E}[\mathbb{I}(A = a)q(A, X, Z)\mathbb{I}(Y \leq y) | U, X = x] \\ = \mathbb{E}[\mathbb{I}(Y \leq y)q(A, X, Z) | A = a, U, X = x]f(A = a | U, X = x) \\ = \mathbb{E}[\mathbb{I}(Y \leq y) | A = a, U, X = x] \\ \times \mathbb{E}[q(A, X, Z) | A = a, U, X = x]f(A = a | U, X = x) \\ = \mathbb{E}[\mathbb{I}(Y \leq y) | A = a, U, X = x] \end{aligned}$$

We utilize the law of total expectation in the first equality. In the second equality, we leverage the conditional independence assumption. Finally, in the last equality, we apply the following equation mentioned in [24]:

$$\mathbb{E}[q(A, X, Z) | A = a, U, X = x] = \frac{1}{f(A = a | U, X = x)}$$

By consistency,

$$\mathbb{E}[\mathbb{I}(Y \leq y) | A = a, U, X = x] = \mathbb{E}[\mathbb{I}(Y(a) \leq y) | U, X = x]$$

Therefore, given that Equation (A1) holds,

$$\begin{aligned} \mathbb{E}[\mathbb{I}(Y(a) \leq y)] &= \mathbb{E}[\mathbb{E}[\mathbb{I}(Y(a) \leq y) | X = x]] \\ &= \mathbb{E}[\mathbb{E}[\mathbb{I}(A = a)q(A, X, Z)\mathbb{I}(Y \leq y) | X = x]] \\ &= \mathbb{E}[\mathbb{I}(A = a)q(A, X, Z)\mathbb{I}(Y \leq y)] \end{aligned}$$

Setting $y = Q_{a,\tau}$ completes the proof. $\square$

Proof of Lemma 1

Proof. Since $\mathcal{P}_k$ is the RKHS, we can rewrite Equation (6) as

$$\begin{aligned} R_{\mathcal{P}_k}^a(q) &= \sup_{p \in \mathcal{P}_k, \|p\| \leq 1} (\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)\langle p, k((W, X), (\cdot, \cdot)) \rangle])^2 \\ &= \sup_{p \in \mathcal{P}_k, \|p\| \leq 1} (\langle p, \mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))] \rangle)^2 \\ &= \|\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))]\|^2 \end{aligned} \quad (\text{A2})$$

In the first equality, we utilize the reproducing property of $\mathcal{P}_k$, and in the final equality, we rely on the fact that $\mathcal{P}_k$ is a vector space. By assumption, $\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))]$ is a Bochner integrable [48]. Hence, we can write Equation (A2) as

$$\begin{aligned} &\|\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))]\|^2 \\ &= \langle \mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))], \\ &\quad \mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))] \rangle \\ &= \mathbb{E}[\langle (\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot)), \\ &\quad \mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))] \rangle] \\ &= \mathbb{E}[\langle (\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot)), \\ &\quad (\mathbb{I}(A' = a)q(A', X', Z') - 1)k((W', X'), (\cdot, \cdot)) \rangle] \\ &= \mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)(\mathbb{I}(A' = a)q(A', X', Z') - 1) \\ &\quad k((W, X), (W', X'))] \end{aligned}$$

$\square$

Proof of Theorem 1

Proof. First, the law of total expectation implies that

$$\begin{aligned} &\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))] \\ &= \mathbb{E}_{(W, X)}[\mathbb{E}_{(A, Z)}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot)) | (W, X)]] \\ &= \mathbb{E}_{(W, X)}[\mathbb{E}_{(A, Z)}[(\mathbb{I}(A = a)q(A, X, Z) - 1 | (W, X))k((W, X), (\cdot, \cdot))]] \end{aligned}$$

By Lemma 1, we know that $R_k^a(q) = \|\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1)k((W, X), (\cdot, \cdot))]\|^2$. As a result, if $\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1 | (w, x))] = 0$ for $\mathbb{P}_{W, X}$ -almost all (w, x) , $R_k^a(q) = 0$.

To show the converse, we assume that $R_k^a(q) = 0$ and rewrite it as

$$R_k^a(q) = \iint_{(\mathcal{W}, \mathcal{X})} g(\xi)k(\xi, \xi')g(\xi')d\xi d\xi' = 0$$

where $\xi = (w, x)$, $\xi' = (w', x')$ and $g(\xi) = \mathbb{E}_{(A, Z)}[(\mathbb{I}(A = a)q(A, X, Z) - 1 | (w, x))p(w, x)]$. Since k is ISPD by assumption, this implies that g is a zero function with respect to $\mathbb{P}_{W, X}$, i.e., $\mathbb{E}[(\mathbb{I}(A = a)q(A, X, Z) - 1 | (w, x))] = 0$ for $\mathbb{P}_{W, X}$ -almost all (w, x) . $\square$

Uniform Convergence of Risk Functionals

The results in this subsection are used to prove the consistency of $\hat{q}_U$ and $\hat{q}_V$.

Lemma 2. (Uniform consistency of $\hat{R}_{k, U}^a(q)$). Assume that $\mathbb{E}[|q(A, X, Z)|] < \infty$, $\mathcal{Q}$ is compact and Assumption 6 holds. Then, the risk $R_{\mathcal{P}_k}^a(q)$ is continuous about $q \in \mathcal{Q}$ and $\sup_{q \in \mathcal{Q}} |\hat{R}_{k, U}^a(q) - R_{\mathcal{P}_k}^a(q)| \xrightarrow{P} 0$.

Proof. We take the notation $t := (a, x, y, z)$, $t' := (a', x', y', z')$ and $H_q(t, t') = (\mathbb{I}(a = a_0)q(a, x, z) - 1)(\mathbb{I}(a' = a_0)q(a', x', z') - 1)k((w, x), (w', x'))$ for some $t, t' \in \mathcal{A} \times \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$. To prove $\hat{R}_{k, U}^a(q)$ converges uniformly to $R_{\mathcal{P}_k}^a(q)$, we need to show:

(a) $H_q(t, t')$ is continuous at each q with probability one.

$$\begin{aligned} |H_q(t, t')| &= \left| \mathbb{I}(a = a_0)q(a, x, z) - 1 \right| \left| \mathbb{I}(a' = a_0)q(a', x', z') - 1 \right| \\ &\quad \left| k((w, x), (w', x')) \right| \\ &\leq \left| \mathbb{I}(a = a_0)q(a, x, z) - 1 \right| \left| \mathbb{I}(a' = a_0)q(a', x', z') - 1 \right| \\ &\quad \left| k((w, x), (w', x')) \right| \\ &\leq (|q(a, x, z)| + 1)(|q(a', x', z')| + 1) \\ &\quad \sqrt{k((w, x), (w, x))k((w', x'), (w', x'))} \equiv d(t, t') \end{aligned}$$

The inequality is obtained using the Cauchy-Schwarz inequality and absolute value inequality. Since $\mathcal{Q}$ is compact, every $q \in \mathcal{Q}$ has $q(a, x, z)$ bounded for $\| (a, x, z) \| < \infty$. In terms of $k(\cdot, \cdot)$ is bounded as per Assumption 6, we have $H_q(t, t') < \infty$ and thus $H_q(t, t')$ continuous at each q with probability one.

(b) There is $d(t, t')$ with $|H_q(t, t')| \leq d(t, t')$ for all $q \in \mathcal{Q}$ and $\mathbb{E}_{T, T'}[d(T, T')] < \infty$.

$$\mathbb{E}_{T, T'}[d(T, T')] \leq \mathbb{E}[1 + |q(A, X, Z)|^2 \sup_{(w, x)} k((w, x), (w, x))] < \infty$$

(c) $(t_i, t_j)_{i \neq j}^{n, n}$ has strict stationarity and ergodicity in the sense of [49].

Strict stationarity means that the distribution of a set of data $(t_i, t_{j \neq i})_{i=1, j=1}^{i+m, j+m'}$ does not depend on the starting indices i, j for any m and m' , which is easy to check. Ergodicity means that $\hat{R}_{k, U}^a(q) \xrightarrow{P} R_{p_k}^a(q)$ for all $q \in \mathcal{Q}$ and $\mathbb{E}[|H_q(T, T')|] < \infty$. We have already shown that $H_q(t, t')$ is bounded and $\mathbb{E}[|H_q(T, T')|] < \infty$, so $\hat{R}_{k, U}^a(q) \xrightarrow{P} R_{p_k}^a(q)$ follows by [50]. Therefore, ergodicity holds, and we have shown all conditions required by extended results of [49] (Lemma 2.4). Then, it follows that $\sup_{q \in \mathcal{Q}} |\hat{R}_{k, U}^a(q) - R_{p_k}^a(q)| \xrightarrow{P} 0$ and $R_{p_k}^a(q)$ is continuous. $\square$

Lemma 3. (Uniform consistency of $\hat{R}_{k, V}^a(q)$). Assume that $\mathbb{E}[\sup_{q \in \mathcal{Q}} |q(A, X, Z)|^2] < \infty$, $\mathcal{Q}$ is compact and Assumption 6 holds. Then, the risk $R_{p_k}^a(q)$ is continuous about $q \in \mathcal{Q}$ and $\sup_{q \in \mathcal{Q}} |\hat{R}_{k, V}^a(q) - R_{p_k}^a(q)| \xrightarrow{P} 0$.

Proof. We adopt the same notation as used in the proof of Lemma 2. To prove that $\hat{R}_{k, V}^a(q)$ converges uniformly to $R_{p_k}^a(q)$, we need to show:

(a) $H_q(t, t')$ is continuous at each q with probability one, as proven in Lemma 2.

(b) $\mathbb{E}_{T, T'}[\sup_{q \in \mathcal{Q}} |H_q(T, T')|] < \infty$ and $\mathbb{E}_{T, T'}[\sup_{q \in \mathcal{Q}} |H_q(T, T')|] < \infty$.

$$\begin{aligned} \mathbb{E}_{T, T'}[\sup_{q \in \mathcal{Q}} |H_q(T, T')|] &\leq \mathbb{E}[\sup_{q \in \mathcal{Q}} (|q(A, X, Z)| + 1)(|q(A', X', Z')| + 1) \\ &\quad \sqrt{k((W, X), (W, X))k((W', X'), (W', X'))}] \\ &\leq \mathbb{E}[\sup_{q \in \mathcal{Q}} (|q(A, X, Z)| + 1) \sup_{q \in \mathcal{Q}} (|q(A', X', Z')| + 1) \\ &\quad \sup_{(w, x)} k((w, x), (w, x))] \\ &= \mathbb{E}[\sup_{q \in \mathcal{Q}} (|q(A, X, Z)| + 1)^2 \sup_{(w, x)} k((w, x), (w, x))] \\ &= \mathbb{E}[\sup_{q \in \mathcal{Q}} (|q(A, X, Z)| + 1)^2 \sup_{(w, x)} k((w, x), (w, x))] < \infty \\ \mathbb{E}_{T, T'}[\sup_{q \in \mathcal{Q}} |H_q(T, T')|] &\leq \mathbb{E}[\sup_{q \in \mathcal{Q}} (|q(A, X, Z)| + 1)^2 \sup_{(w, x)} k((w, x), (w, x))] \\ &= \mathbb{E}[\sup_{q \in \mathcal{Q}} (|q(A, X, Z)| + 1)^2 \sup_{(w, x)} k((w, x), (w, x))] \\ &\leq 2 \left(\mathbb{E}[\sup_{q \in \mathcal{Q}} |q(A, X, Z)|^2] + 1 \right) \sup_{(w, x)} k((w, x), (w, x)) < \infty \end{aligned}$$

Hence, our assertion follows from [49] (Lemma 8.5). $\square$

Proof of Theorem 2

Proof. From the conditions of Lemmas 2 and 3, we know that $\mathcal{Q}$ is compact, $R_{p_k}^a(q)$ is continuous about q and $\sup_{q \in \mathcal{Q}} |\hat{R}_{k, U(V)}^a(q) - R_{p_k}^a(q)| \xrightarrow{P} 0$. As $\mathcal{Q}$ is convex and Assumptions 5 and 6 hold, $R_{p_k}^a(q)$ is convex on $\mathcal{Q}$ and is uniquely minimized at q^* . Based on the conditions that $\Lambda(q)$ is bounded and $\lambda \xrightarrow{P} 0$, we obtain by Slutsky's theorem that

$$\begin{aligned} \sup_{q \in \mathcal{Q}} |\hat{R}_{k, U(V)}^a(q) + \lambda \Lambda(q) - R_{p_k}^a(q)| &\leq \sup_{q \in \mathcal{Q}} |\hat{R}_{k, U(V)}^a(q) - R_{p_k}^a(q)| \\ &\quad + \lambda \sup_{q \in \mathcal{Q}} \Lambda(q) \xrightarrow{P} 0 \end{aligned}$$

Consequently, we assert the conclusion by [49] (Theorem 2.1). $\square$

Proof of Theorem 3

Proof. The difference between $\hat{\varphi}_{U(V)}$ and φ can be decomposed as

$$\begin{aligned} \hat{\varphi}_{U(V)} - \varphi &= \underbrace{\left(\frac{1}{n} \sum_{i=1}^n (-1)^{1-a_i} (\hat{q}_{U(V)}(a_i, x_i, z_i) - q^*(a_i, x_i, z_i)) y_i \right)}_{(I)} \\ &\quad + \underbrace{\left(\frac{1}{n} \sum_{i=1}^n (-1)^{1-a_i} q^*(a_i, x_i, z_i) y_i - \varphi \right)}_{(II)} \end{aligned}$$

To prove $\hat{\varphi}_{U(V)} \xrightarrow{P} \varphi$, we need to show both terms (I) and (II) converge to 0 in probability.

Term (I): By the consistency of $\hat{q}_{U(V)}$, $\sup_i |\hat{q}_{U(V)}(a_i, x_i, z_i) - q^*(a_i, x_i, z_i)| \xrightarrow{P} 0$. Since Y and $(-1)^{1-a_i}$ are bounded, this term converges to 0 in probability.

Term (II): This term represents the discrepancy between the empirical average and the true expectation:

$$\frac{1}{n} \sum_{i=1}^n (-1)^{1-a_i} q^*(a_i, x_i, z_i) y_i - \mathbb{E}[(-1)^{1-A} q^*(A, X, Z) Y]$$

By the law of large numbers, the empirical average converges to its true expectation, implying that this term converges to 0 in probability. $\square$

Appendix B

Experiment Settings

Covariates X are generated from a multivariate normal distribution $N(\Gamma_x, \Sigma_x)$. We then generate A conditional on X from a Bernoulli distribution.

Next we generate Z, W, U from the following multivariate normal distribution,

$$(Z, W, U) | A, X \sim MVN \left(\begin{pmatrix} \alpha_0 + \alpha_a A + \alpha_x X \\ \mu_0 + \mu_a A + \mu_x X \\ \kappa_0 + \kappa_a A + \kappa_x X \end{pmatrix}, \Sigma = \begin{pmatrix} \sigma_z^2 & \sigma_{zw} & \sigma_{zu} \\ \sigma_{zw} & \sigma_w^2 & \sigma_{wu} \\ \sigma_{zu} & \sigma_{wu} & \sigma_u^2 \end{pmatrix} \right)$$

Finally, Y is generated from $\mathbb{E}[Y|A, U, W, X, Z]$ plus a normal noise $N(0, \sigma_y^2)$ with

$$\begin{aligned} \mathbb{E}[Y|A, U, W, X, Z] &= b_0 + b_1(X)A + b_2(X)X \\ &\quad + (b_w + b_a A + b_3(X)A - \omega) \mathbb{E}[W|U, X] + \omega W \end{aligned}$$

where $\mathbb{E}[W|U, X] = \mu_0 + \mu_x X + \frac{\sigma_{wu}}{\sigma_u^2} (U - \kappa_0 - \kappa_x X)$.

The parameters are set as follows:

$$\begin{aligned}\Gamma_x &= (0.25, 0.25)^T, \Sigma_x = \begin{pmatrix} 0.625 & 0 \\ 0 & 0.625 \end{pmatrix} \\ \Pr(A = 1|X) &= [1 + \exp \{ (0.25, 0.25)^T X \}]^{-1} \\ \alpha_0 &= 0.25, \alpha_a = 0.25, \alpha_x = (0.25, 0.25)^T \\ \mu_0 &= 0.25, \mu_a = 0.25, \mu_x = (0.25, 0.25)^T \\ \kappa_0 &= 0.25, \kappa_a = 0.25, \kappa_x = (0.25, 0.25)^T \\ \Sigma &= \begin{pmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 1 \end{pmatrix}, \sigma_y = 0.25 \\ b_0 &= 2, \omega = 1\end{aligned}$$

The remaining parameters $b_1(X)$, $b_2(X)$, $b_3(X)$, b_a , and b_w are described in Table 1.

Appendix C

Hyperparameter Tuning

We performed hyperparameter tuning using the `gp_minimize` function from the `scikit-optimize` Python library. The search space for the hyperparameters of the neural network is summarized in Table C1.

TABLE C1 | Hyperparameter search space for simulation and RHC. Int denotes integer variables, and Real denotes continuous variables.

	Network depth	Network width	Learning rate	L_2 penalty	Dropout prob
Simulation	Int(3, 6)	Int(20, 40)	Real (10^{-5} , 10^{-2})	Real (10^{-8} , 10^{-7})	Real (0.1, 0.5)
RHC	Int(4, 9)	Int(20, 50)	Real (10^{-5} , 10^{-2})	Real (10^{-8} , 10^{-7})	Real (0.1, 0.5)

For each experiment, we selected the optimal hyperparameters by minimizing the value of the regularized loss function. Another important hyperparameter is the kernel used in the loss function of DeepMMR. In all experiments, we employed the RBF kernel.

$$K(x_i, x_j) = \exp \{ -\gamma^{-1} \|x_i - x_j\|_2^2 \}$$

where $\gamma = \text{median} \{ \|x_i - x_j\|_2^2 : 1 \leq i < j \leq n \}$ is the median heuristic bandwidth parameter.

Appendix D

Experiment Results

Simulation results corresponding to sample sizes $n = 2000$ and $n = 4000$ are summarized in Tables D1 and D2.

TABLE D1 | Simulation results ($n = 2000$): Means and standard deviations.

	PMMR	GMM	MinMax	DeepMMR-U	DeepMMR-V
S1	1.236 (0.092)	0.916 (0.264)	0.983 (0.252)	1.008 (0.235)	0.927 (0.156)
S2	1.221 (0.105)	0.911 (0.251)	0.962 (0.279)	0.987 (0.176)	1.021 (0.162)
S3	1.226 (0.091)	0.887 (0.277)	0.986 (0.251)	0.955 (0.164)	1.006 (0.164)
S4	1.228 (0.076)	0.888 (0.222)	0.980 (0.223)	0.962 (0.142)	0.988 (0.141)
S5	1.192 (0.088)	0.894 (0.245)	0.974 (0.216)	0.997 (0.164)	0.969 (0.141)
S6	1.193 (0.098)	0.855 (0.229)	0.961 (0.238)	1.015 (0.156)	1.007 (0.159)

TABLE D2 | Simulation results ($n = 4000$): Means and standard deviations.

	PMMR	GMM	MinMax	DeepMMR-U	DeepMMR-V
S1	1.306 (0.088)	0.898 (0.221)	1.042 (0.301)	1.014 (0.154)	1.013 (0.137)
S2	1.277 (0.079)	0.923 (0.255)	1.013 (0.293)	1.001 (0.147)	1.047 (0.145)
S3	1.278 (0.085)	0.895 (0.242)	1.063 (0.307)	1.010 (0.146)	0.970 (0.138)
S4	1.292 (0.071)	0.937 (0.214)	1.030 (0.262)	1.007 (0.115)	0.988 (0.127)
S5	1.260 (0.079)	0.892 (0.257)	1.007 (0.263)	0.977 (0.114)	1.002 (0.124)
S6	1.267 (0.086)	0.863 (0.215)	0.995 (0.279)	1.004 (0.132)	1.003 (0.128)

Appendix E

Mini-Batch Training

The primary computational bottleneck of our approach arises from constructing the kernel matrix, which entails both time and space complexities of $\mathcal{O}(n^2)$ with respect to the sample size n , thereby limiting its scalability to large datasets. To address this issue, we employ a mini-batch training scheme aimed at reducing memory consumption, enhancing computational efficiency, and stabilizing the training process.

Specifically, we apply the mini-batch strategy during the training of the deep neural network q . A PyTorch `DataLoader` is used to randomly sample mini-batches of size 100 from the training set. At each iteration, the kernel matrix and loss function are computed based on the current mini-batch, and the model parameters are updated via stochastic gradient descent by minimizing this mini-batch approximation of the loss.

To assess the impact of the mini-batch strategy, we conduct experiments comparing model performance with and without mini-batching across varying sample sizes. The results, summarized in Table E1, report average running times over 100 runs for six scenarios, along with mean CPU and memory usage recorded during the experiments. The findings demonstrate that the mini-batch strategy significantly enhances training efficiency and scalability, with pronounced benefits for larger datasets.

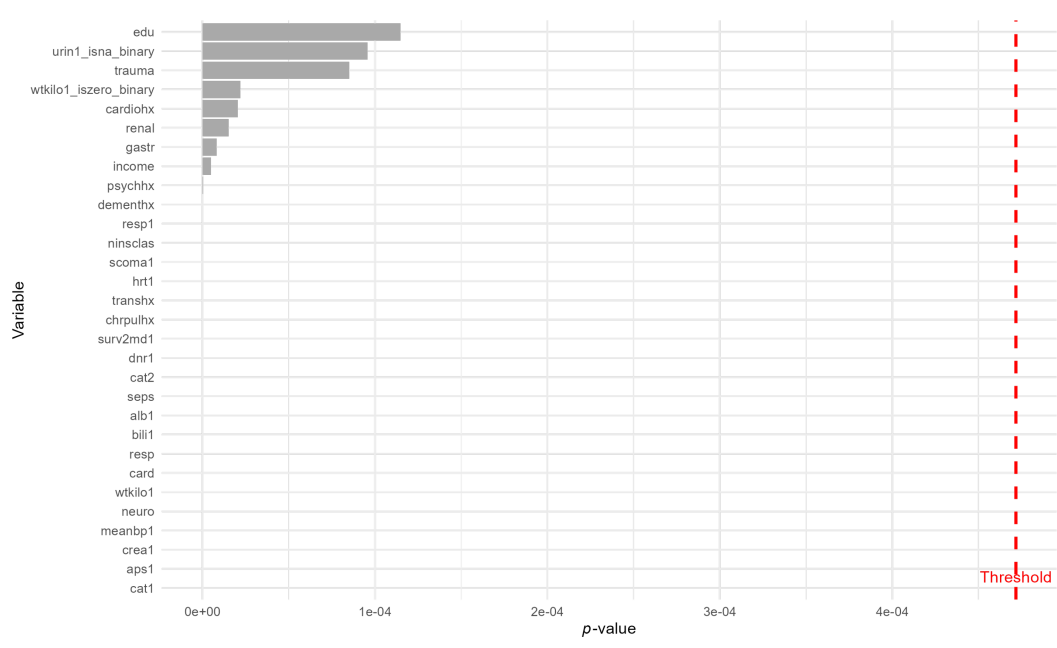
TABLE E1 | Comparison of model performance with and without mini-batching across varying sample sizes.

Sample size	Batch strategy	Running time (min)	CPU usage (%)	Memory (GB)
$n = 2000$	With mini-batch	20.86	8.81	0.35
	Without mini-batch	71.71	8.87	0.47
$n = 4000$	With mini-batch	50.77	8.83	0.36
	Without mini-batch	306.67	9.30	0.66

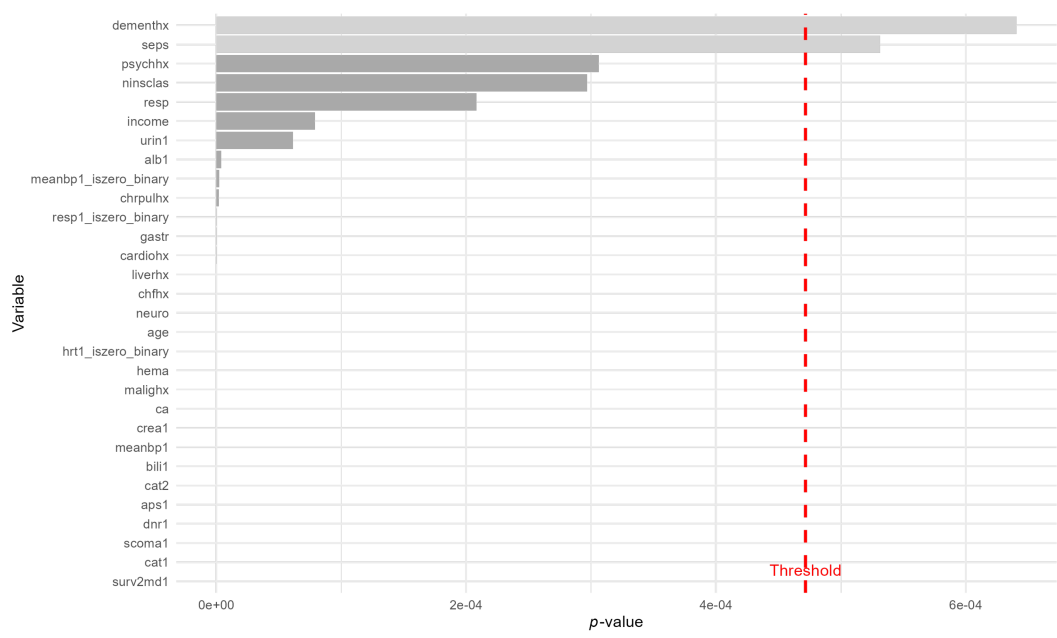
Appendix F

RHC Data Application

To assess the significance of variables in the dataset, we employed appropriate statistical tests based on variable types to obtain corresponding p -values with respect to the treatment A and the outcome Y . Specifically, we used the Chi-squared test for categorical and binary variables, and the Mann-Whitney U test for continuous variables to evaluate their relationships with the treatment A . For associations with the outcome Y , we utilized the Kruskal-Wallis H test for categorical variables, the Mann-Whitney U test for binary variables, and Spearman correlation for continuous variables. To reduce the false discovery rate, we applied the Bonferroni correction to adjust the significance threshold. Figure F1 demonstrates the results of the statistical tests.



(a) p -value with the treatment A



(b) p -value with the outcome Y

FIGURE F1 | Significant variables selected by p -values.

Based on statistical test results, we assigned 20 significant variables to X . These variables are categorized as follows: categorical variables (cat1, cat2, scoma1, ninsclas, income), binary variables (cardiohx, psychhx, chrpulhx, malighx, dnr1, resp., neuro, gastr), and continuous variables (meanbp1_iszero_binary, surv2md1, alb1, bili1, meanbp1, crea1, aps1).